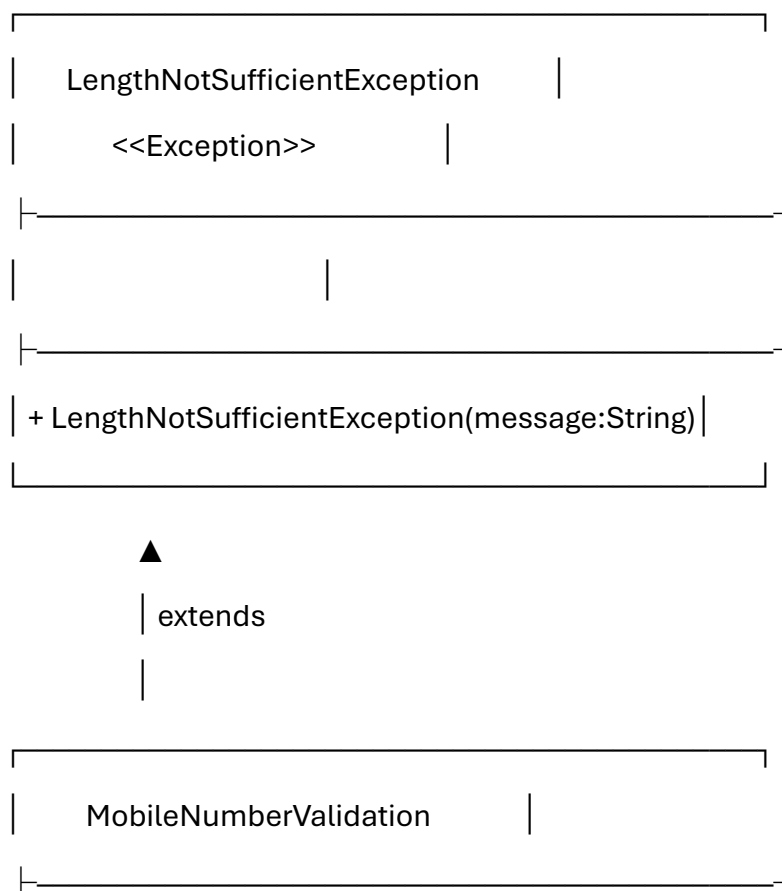


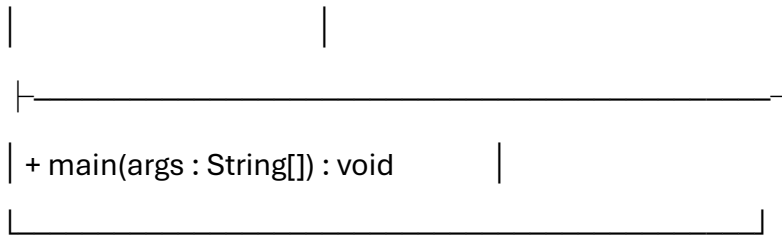
To write a Java program to validate a mobile number using **exception handling** and demonstrate **ArrayIndexOutOfBoundsException**, **user-defined LengthNotSufficientException**, and **NumberFormatException**.

Algorithm

1. Read the mobile number as a string.
2. Check each character of the mobile number.
3. If any character is not a digit, throw **NumberFormatException**.
4. If the length is greater than 10, access an invalid array index to generate **ArrayIndexOutOfBoundsException**.
5. If the length is less than 10, throw the user-defined **LengthNotSufficientException**.
6. If the length is exactly 10 and all characters are digits, display **Valid number**.
7. Handle all exceptions using multiple catch blocks.
8. Display the appropriate error message

UML Diagram:





Code:

```
package Records;

import java.util.Scanner;

// User-defined exception
class LengthNotSufficientException extends Exception {

    LengthNotSufficientException(String message) {

        super(message);

    }

}

public class ExpectionHandling {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter mobile number: ");

        String mobile = sc.nextLine();

        try {

            // Check whether all characters are digits

            for (int i = 0; i < mobile.length(); i++) {
```

```
        if (!Character.isDigit(mobile.charAt(i))) {  
            throw new NumberFormatException();  
        }  
    }  
}
```

```
// If length is greater than 10
```

```
if (mobile.length() > 10) {  
    int[] arr = new int[10];  
    System.out.println(arr[mobile.length()]);  
}
```

```
// If length is less than 10
```

```
if (mobile.length() < 10) {  
    throw new LengthNotSufficientException(  
        "Invalid Mobile Number - LengthNotSufficientException"  
    );  
}
```

```
// Valid number
```

```
System.out.println("Valid number");
```

```
} catch (ArrayIndexOutOfBoundsException e) {
```

```
    System.out.println(  
        "Invalid Mobile Number - ArrayIndexOutOfBoundsException"  
    );  
}
```

```
} catch (LengthNotSufficientException e) {
```

```
    System.out.println(e.getMessage());  
}
```

```

    } catch (NumberFormatException e) {
        System.out.println(
            "Invalid Mobile Number - NumberFormatException"
        );
    }

    } finally {
        sc.close();
    }
}
}

```

Output:

```

Records.ExceptionHandling'
Enter mobile number: 9182567489
Valid number
PS C:\Users\Sahithi\pyla\Desktop\Sahithi_java> ^C
PS C:\Users\Sahithi\pyla\Desktop\Sahithi_java>
PS C:\Users\Sahithi\pyla\Desktop\Sahithi_java> c::; cd '
nable-preview' '-XX:+ShowCodeDetailsInExceptionMessages'
5cfd\redhat.java\jdt_ws\Sahithi_java_e658ba66\bin' 'Reco
Enter mobile number: 00917375
Invalid Mobile Number - LengthNotSufficientException
PS C:\Users\Sahithi\pyla\Desktop\Sahithi_java>

```